

of computer-controlled robots used to assist humans.

As an electronics professional, you may find a place in the design section of a robot manufacturing firm. Fresh electronics graduates can be considered for some of the positions in the creative designing and problem-solving sections. They may have to work on electronic designs and embedded programs for robotic assemblies like actuators, sensors and interface circuitries and develop innovative graphical user interfaces (GUIs) and control mechanisms for the robots.

In the manufacturing section, graduates may play important roles in maintenance and operation. Microchip manufacturers recruit robotics and artificial intelligence specialists. Employment opportunities also exist in education and research domains. Electronics departments in most of the IITs and CSIR labs offer research fellowships in robotics and artificial intelligence.

As a robotics engineer, you may find placement in robot programming, troubleshooting and maintenance. Fellowships are also available in research institutes, emphasising on humanoid robotics and computational neuroscience.

Those with overall knowledge in this area may work as CAD/CAM engineers, automation engineers, hydraulics and pneumatic system engineers, engineers for numerically controlled systems and microprocessors, robotic system programmers and maintenance engineers.

The skills and knowledge of these engineers may be used to design robots to carry out hazardous or repetitive tasks in mining or chemical waste disposal. For instance, robots are being deployed at the Fukushima Daiichi nuclear power plant in Japan which was recently damaged by a massive earthquake and tsunami and is still too dangerous for humans to tread.

These engineers may also be employed to create life-saving surgical equipment; help design advanced toys, games and household appliances for the commercial industry; or develop

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experimental robots in the research sector for use in defence, surveillance and aerospace programmes.

Apart from this, there are various opportunities in procurement, manufacturing and technical sales in robotic companies.

How to get the entry?

A technical degree (diploma or graduation) may qualify you for a job in this field but it may not be enough to get you a decent profile. As recruiters also look for specialised R&D skills in a postgraduate or doctorate applicant, you need to have a knack for the field. Most employers train interns directly in various areas of robotics.

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Courses in robotics are offered by institutes like IIT-Kanpur; National Institute of Technology, University of Hyderabad; Kuka Training Centre, Pune; Jadavpur University, Kolkata; BITS Pilani; Sri Sathya Sai Institute of Higher Learning, Prasanthinilayam in Tamil Nadu; and PSG College of Technology, Coimbatore.

Money matters

"The remuneration in robotics is at par with any other top-rated field like aerospace and networking. At the same time, the matrix for fixing the salary for a particular position follows the same

pattern as for any other IT-related position to be mutually beneficial for the employer and the employee," says Jayakrishnan.

"The difference is much higher for an experienced candidate as it is not very easy to get the right combination to match the position. Anyhow, due to the technical content of the job, the package is definitely on the higher side as far as IT jobs are concerned," he adds.

Mel Siegel, an associate research professor in the Robotics Institute at Carnegie Mellon University (CMU), shares the same view: "As in other technical fields, salaries are good even for beginners in technical jobs, becoming excellent as people move into managerial jobs. A few of our graduates became quite wealthy when their entrepreneurial activities culminated in selling the small companies that they founded to large companies that wanted their product lines and expertise."

Apart from graduate professionals, the field offers a decent starting salary in the range of Rs 10,000 to 25,000 per month to diploma holders.

Major players

Most recruiters are from the automation industry, doling out intelligent systems and autonomous systems involving highly advanced engineering and technology. Most of their products are for military and defence applications.

Automobile companies account for 70 per cent of robotics applications in India. Tata Motors, Hyundai Motors, Reliance Industries, Mahindra & Mahindra and Ashok Leyland have